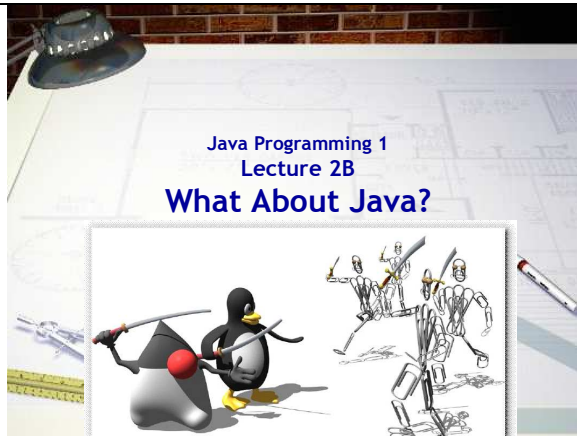




Slide 1
Java Programming 1
Lecture 2B
What About Java?
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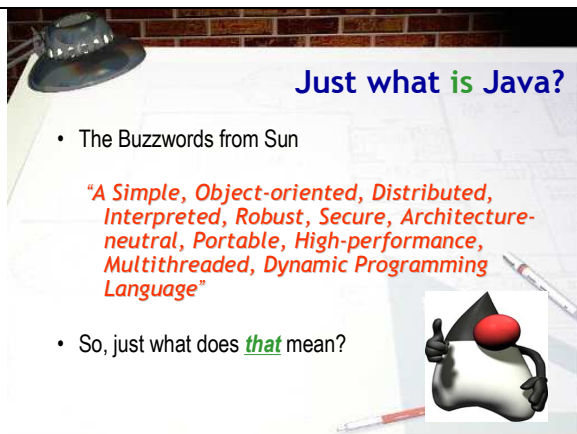
Notes:

After looking at computer memory, the CPU, and a little computing history, we finally, (I hear some of you say), come to Java. Java is kind of a controversial topic in the computing world; Sun Microsystems and Microsoft have exchanged lawsuits over Java, and Java figured prominently in the government's Microsoft antitrust trial. If you're like most people, you've probably asked yourself, "Why?"

Let's start by taking a "high-level" look at Java. Part of the controversy surrounding Java occurs because different people mean different things when they talk about Java. In this lecture I'll try to answer these three questions about Java:

- what is meant by "the Java platform"
- why Java is important
- why Computer Science students should learn Java

Let's start by answering the question, "What is Java?"



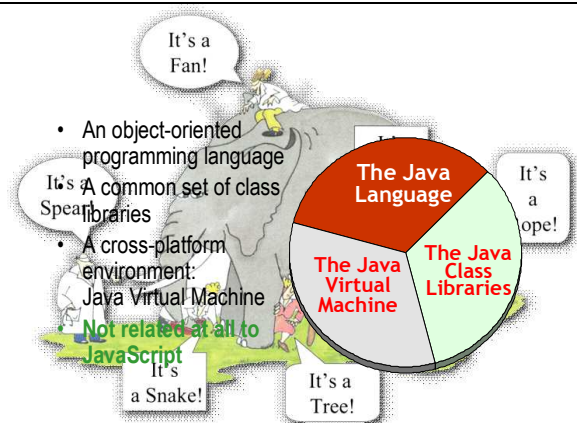
Slide 2
Just what is Java?
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Notes:

Java is a confused and misused term in the field of computing. Sun Microsystems, the developers of Java, hoping to alleviate some of this confusion, have provided a simple definition; according to Sun, Java is :

"A Simple, Object-oriented, Distributed, Interpreted, Robust, Secure, Architecture-neutral, Portable, High-performance, Multithreaded, Dynamic Programming Language"

Certainly nothing controversial there! (I'm sure that all of you find Sun's explanation just as illuminating as I do :)! Instead of walking through each of these "buzzwords", let's step back and take a look at what I call the three faces of Java.



Slide 3

Notes:

A well-known Indian legend tells the story of seven blind men describing an elephant. The one who felt the trunk thought the elephant was like a snake, the one who felt the legs thought the elephant was like a tree, and so on.

Java is a little bit like that; you'll often hear people say things like:

- "Java is slow" or, less often, "Not so! Java is just as fast as C++"
- Java programs look "strange" when I run them
- Java is better for building Web applications than

The “Java Platform”

Duration: 00:02:40

Advance mode: Auto

Microsoft .NET

- Microsoft .NET supports Java along with other programming languages
- Java is cross-platform while VB-script only works in Internet Explorer

Each of these statements, (except the last), contains a kernel of truth; each also suffers from confusion between the three parts of the Java platform. In reality, Java is:

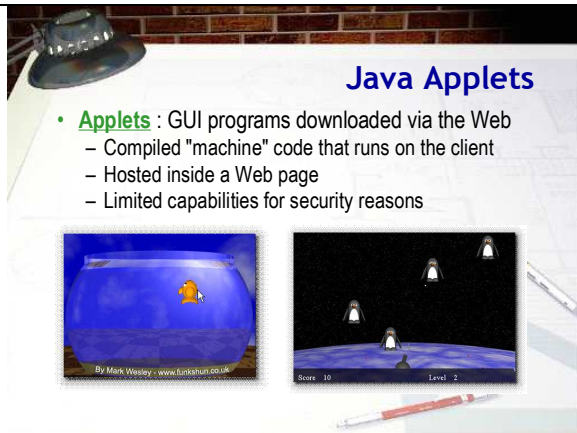
- an object-oriented programming language. The Java language, like C++ or SmallTalk, can be supported on different runtime systems, like the Sun Virtual Machine, or even on Microsoft's .NET.
- a set of universally distributed pre-built programming components called the Java Class Libraries. These class libraries provide support for GUI programming, network and file I/O, etc. that you can use when building your Java programs.
- a runtime system called the Java Virtual Machine or JVM. There are several different kinds of JVM's. Some use an interpreter, some use a Just In Time (JIT) compiler. All of them, though, use a "virtual" machine language called bytecode.

The combination of all three of these is called a Java platform. The most commonly used Java Platform is supplied by Sun (called the JRE), but there are also versions supplied by Apple, Microsoft, IBM, HP, and so on.

JavaScript is often confused with Java, but the two are really not related at all. JavaScript is a programming language designed by Netscape, used to manipulate Web documents and Web browsers. JavaScript was originally named LiveScript, but was renamed by Netscape when Java became popular.

JavaScript programs are downloaded to your Web browser, often embedded inside an HTML program, and the instructions are interpreted and executed by your Web browser. JavaScript and Java are two entirely different languages, used for different purposes.

So, if Java isn't the same as JavaScript, then what is Java used for? What kind of programs are written in Java? Let's look at that next.



Java Applets

- **Applets** : GUI programs downloaded via the Web
 - Compiled "machine" code that runs on the client
 - Hosted inside a Web page
 - Limited capabilities for security reasons

Slide 4

Java Applets

Duration: 00:01:19

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Notes:

There are four major types of Java programs:

- applets,
- applications,
- server-side programs,
- and mobile applications.

Let's take a look at each of these.

Java applets are programs that are "hosted" inside a Web page. Here's how Java applets work:

- The compiled machine code for your program is stored on a Web server, much like an .gif or .jpg image file, or like a Flash animation file.
- The code is automatically downloaded to the viewer's machine when a user visits a Web page that contains a reference to the applet (called an applet tag).
- The user's Web browser starts running a Java Virtual Machine when it encounters a Java applet, and the JVM runs the applet inside the browser's JVM. This is similar to the way Flash/Shockwave animation, or other browser plug-ins, like QuickTime, are handled.
- For safety sake, Java applets are prohibited from performing actions that might damage a user's machine if used maliciously. Applets cannot read or write files on the user's local machine, for instance.

Sometimes, you'll hear applets referred to as "client-side" Java. In this class, you'll learn to create both applets, and a second kind of Java program, applications.

Slide 5

Java Applications

Duration: 00:02:04

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Notes:

Java applications are programs that are installed and run on the user's local machine, in the same way you would install and run a program written in Visual Basic or C++. Java applications are not inherently safe like Java applets are; there are no built-in security limitations so you should follow the same cautions with Java applications you'd follow when you install any kind of software.

There are two kinds of Java applications:

- Console-mode applications are traditional "teletype-style" programs similar to those you'd write in a Pascal or C++ class. Many of the programs you'll write in this class will be console applications. Console programs are useful when you're learning to program because they reduce the amount of detail that can "clutter up" what you're trying to focus on. Console programs are also often used for utility and server-based applications.
- GUI (Graphical User Interface) applications. These

programs use windows, buttons, checkboxes and graphics to create the kind of interactive programs you're used to in the Windows & Mac worlds.

Java actually contains two different GUI libraries, the original AWT (Abstract Window Toolkit) which uses your operating system's native look-and-feel, and the JFC (Java Foundation Classes--AKA Swing), which provides a set of advanced components that can be made to look and feel the same on all platforms. Your textbook uses the JFC components, but I'll try to cover the AWT classes as well in our lectures. (The AWT is useful for writing applets, because earlier Java-enabled Web browsers don't have Swing installed.)

Other GUI libraries, like the Standard Widget Toolkit (SWT) from IBM, (used to create the Eclipse IDE) are available, but are not part of the standard, universal Java class libraries. During a portion of the class, we'll also be using a library of graphical shapes developed by the Association of Computing Machinery or ACM.



Slide 6

Server and Mobile Apps

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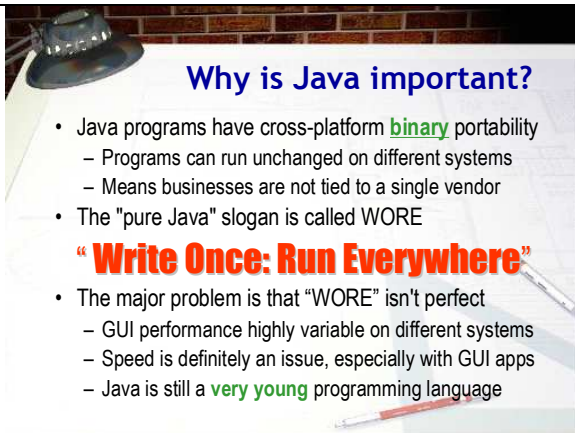
Notes:

A third kind of Java program is called a server-side application. These programs use an additional set of Java classes, in addition to those contained in the Java 2 Standard Edition (J2SE). Collectively, these additional classes are called the Java 2 Enterprise Edition or J2EE. J2EE does not replace standard Java, which you'll learn in this class, but adds new features to it.

Server-side applications are not downloaded or installed on the user's computer. Instead, the Java program runs on an Application Server, (such as BEA WebLogic, IBM WebSphere, or Apache's Tomcat), and only the input or output is sent to the user's machine, as regular HTML.

Server-side applications are most often accessed through a Web browser, and are used for ecommerce and other interactive Web sites. We won't do any server-side Java in this class. OCC's Java Programming 2 course, CS 272, includes server-side Java programming along with advanced GUI and database programming.

You can also write Java programs that will run on Palm Pilots, cell phones, and other "low resource" devices. These programs use the same Java language, but an entirely different set of class libraries. The class libraries used for mobile devices are called the Java 2 Micro Edition or J2ME.



Why is Java important?

- Java programs have cross-platform **binary** portability
 - Programs can run unchanged on different systems
 - Means businesses are not tied to a single vendor
- The "pure Java" slogan is called WORE

"Write Once: Run Everywhere"

- The major problem is that "WORE" isn't perfect
 - GUI performance highly variable on different systems
 - Speed is definitely an issue, especially with GUI apps
 - Java is still a **very young** programming language

Slide 7

Why is Java important?

Duration: 00:02:30

Advance mode: Auto

Notes:

Java is a portable language. A portable language is one that is designed to run on different computer operating systems or different computer hardware. The C language is portable, for instance, because you can run the same program on both your Windows computer and your Macintosh, by taking your C-language source code and recompiling it with a different C compiler on each machine. JavaScript is also a portable language, because you can download the same program to a Mac user, a Windows user, or a Unix user, and the Web browser will interpret the source code, turning it into the correct machine code for the individual machine.

This is called source code portability. Java programs are designed to offer more than source-code portability. With Java programs, you can actually take your binary bytecode, compiled on a Mac, and then run it, with no changes at all, on a Windows or Unix computer. This is possible because the JVM installed on each platform understands the same bytecode, and because every Java platform has the same classes available. This means that businesses can develop their programs on inexpensive Windows desktop machines and then move them to larger Sun servers or even IBM mainframes as their business grows.

Sun's slogan for the Java Platform is summed up in:

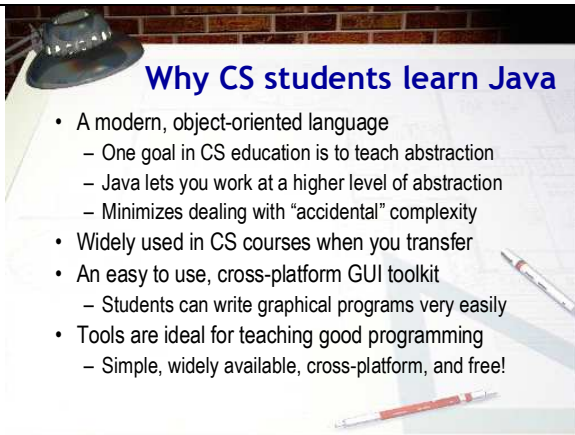
"Write Once: Run Everywhere"

WORE is designed to allow your programs to run unchanged on any operating system or any hardware that has a JVM. Of course that's also a major weakness with Java: WORE really isn't perfect; each implementation of the Java platform has subtle differences, and so developers often end up with:

"Write Once: Test Everywhere"

There are several reasons for this:

- GUI performance depends a lot on the native platform. In other words, Button objects on the Mac act differently from Button objects on Windows or Unix, and it's difficult for Java to emulate that exactly. It's kind of a moving target.
- Java's speed is still definitely an issue. Java GUI apps use more memory and perform more slowly than equivalent C++ programs. (Of course, that is also true about Microsoft's latest languages, like C# and VB.NET as well.)
- *Finally, Java is still a pretty young language (first released in 1997), so it is still going through a lot of changes. (Some folks would say too many changes.)

The slide features a background image of a desk with a lamp, a pen, and some papers. The title "Why CS students learn Java" is in blue. The content is a bulleted list.

Why CS students learn Java

- A modern, object-oriented language
 - One goal in CS education is to teach abstraction
 - Java lets you work at a higher level of abstraction
 - Minimizes dealing with "accidental" complexity
- Widely used in CS courses when you transfer
- An easy to use, cross-platform GUI toolkit
 - Students can write graphical programs very easily
- Tools are ideal for teaching good programming
 - Simple, widely available, cross-platform, and free!

Slide 8

Why CS students learn Java

Duration: 00:02:06

Advance mode: Auto

Notes:

If Java programs are less capable than C++ programs, and they run slower, then why not just learn C++ to start with, instead of messing with Java? First, Java is a modern, object-oriented language. One of the major goals of CS education is to teach you to use abstraction. Abstraction is the ability to mentally "strip away" the non-essential details in a problem, and concentrate on the core elements to craft your solution, and Java lets you work at a higher level of abstraction than C or C++, or even C#. While there are still a lot of details you have to get exactly right, compared to C++ the amount of extraneous minutia has been considerably reduced.

Java is also used extensively in Computer Science classes at Universities across the country, not only for the introductory programming classes (like this one) but for your upper-division coursework as well. If you plan on transferring, your life will be a lot easier if you know Java well.

Another nice bonus for CS students is that you can easily write graphical applications and, you can do your work on the Mac, Linux or on Windows: you choose. By contrast, writing graphical applications in C++ is much more complex, much too complex for an introductory programming class. Writing the "Hello World" application for Microsoft Windows in C++, for instance, takes a full page of code and involves pointers, structures static variables and callbacks.

Finally, the tools that we'll use in this course are ideal for teaching programming. They're generally simple and easy to learn to use (although we will be using a more complex, professional IDE called Eclipse during the second half of the class). All of the tools we'll use are widely available via the Internet and are all cross-platform. Finally, all of the tools that we'll use are free (which I'm sure you'll agree is important).